

Remark 4.3. Recall the notation S_j for the space of functions that are, locally on atoms of $\mathcal{F}_j$, given by functions contained in S . Moreover, P_j denotes the orthoprojector onto S_j . Corresponding to the local orthonormal basis Φ , it is natural to consider weak greedy algorithms corresponding to whole blocks $(P_j - P_{j-1})f$ instead of projections onto single functions $\langle f, \varphi_n \rangle \varphi_n$ for $\varphi_n \in \Phi$.

To wit, let Γ_j be the set of indices n such that φ_n is contained in the range of $W_j = P_j - P_{j-1}$. Consider $\mu_j(f) = \max_{n \in \Gamma_j} |\langle f, \varphi_n \rangle|$ and choose a parameter $0 < t \leq 1$. Consider a weak greedy algorithm given by a sequence (j_k) such that

$$\min_{1 \leq \ell \leq k} \mu_{j_\ell}(f) \geq t \max_{\ell \geq k+1} \mu_{j_\ell}(f)$$

and set

$$B_k f = \sum_{\ell=1}^k W_{j_\ell} f.$$

The results of Lemma 4.1 and Corollary 4.2 then remain valid if $\mathcal{G}_n f$ is replaced by $B_n f$. For a proof of this fact, we refer to Appendix D.

We don't know whether the result of Corollary 4.2 extends also to the variation spaces $V_{\sigma,p}$ instead of approximation spaces, i.e. we don't know if we have the uniform equivalence

$$(4.3) \quad K(f, 2^{-n\alpha}, L^p(S), V_{p,\sigma}) \simeq \|f - \mathcal{G}_{2^n} f\|_p + 2^{-n\alpha} \|\mathcal{G}_{2^n} f\|_{V_{\sigma,p}}$$

for all n . For this, it would suffice that greedy approximation is bounded in variation space, i.e. $\|\mathcal{G}_n f\|_{V_{\sigma,p}} \lesssim \|f\|_{V_{\sigma,p}}$, cf. Lemma 4.1 with the choices $\mathcal{X} = L^p(S)$, $\mathcal{Y} = V_{\sigma,p}$, and $\Psi = \Phi$. Indeed, we will see in (4.9) that (1) of Lemma 4.1 is true and Proposition 3.3 (ii) implies (2) of Lemma 4.1. Therefore, in order to show (4.3), it suffices to show (3) of Lemma 4.1.

On the other hand, we next prove some embeddings between the variation spaces $V_{\sigma,p}$ and the approximation spaces $\mathcal{A}_q^\alpha$ (cf. Sections 4.1 and 4.2) that allows us to conclude a weak version of Corollary 4.2 for variation spaces. This result is given in Theorem 4.6. In the embedding result of Section 4.2 (and therefore also in Theorem 4.6), it is crucial to assume condition $w2^*$ from (2.4) on the geometry of the atoms $\mathcal{A}$ and the space S . Finally, Proposition 4.8 provides an example showing that boundedness of greedy approximation in variation space is not true unless we assume a condition similar to $w2^*$. This suggests that such a geometric condition is somewhat natural in the above embedding results.

4.1. The embedding of $\mathcal{A}_q^\alpha$ into $V_{\sigma,p}$. We have the following result:

Theorem 4.4. Let $1 < p < \infty$, $0 < \sigma < p$ and set $\alpha = 1/\sigma - 1/p$. Letting $q = \min(\sigma, 2)$, the approximation space $\mathcal{A}_q^\alpha$ embeds continuously into the variation space $V_{\sigma,p}$.

In the proof we need the concept of (Rademacher) type ρ (see for instance [12]). We recall that a Banach space $\mathcal{X}$ is of type ρ , $1 \leq \rho \leq 2$, if there exists a constant T such that

$$(4.4) \quad \text{Ave}_\varepsilon \left\| \sum_i \varepsilon_i x_i \right\| \leq T \left(\sum_i \|x_i\|^\rho \right)^{1/\rho}, \quad x_i \in \mathcal{X},$$

where the average is taken over all choices of signs $\varepsilon_i \in \{\pm 1\}$. It is known that L^p is of type $\min(p, 2)$ and clearly, inequality (4.4) for $\rho = r_1$ implies inequality (4.4) for all $\rho = r_2 < r_1$.